Research Highlights (Manuscript ID: Fusion-Arch-001)

1. Compact high-field tokamak architecture verified at B0 = 12.2 T and Ip = 8.54 MA. *[81 chars]*

2. REBCO TF coil loop energy density reaches 59.18 MJ/m3 under 3D structural yield. *[80 chars]*

3. Liquid metal CPS vapor cushion prevents dryout up to 22.50 MW/m2 thermal limit. *[79 chars]*

4. DCLL blanket MHD Shercliff side-wall jet profiles resolved at Ha = 10,000. *[74 chars]*

5. Tsai-Wu structural reserve ratio SRR = 1.61 verified under disruption halo shear. *[81 chars]*